

PRMS ARCHITECTURE LAB

Research Agenda — 2026

Version 1.0 | 2026

Purpose

This agenda defines the current research directions of PRMS Architecture Lab and provides a common structure for projects, publications, prototypes, measurements, and future research collaborations.

Central Research Question

How can architecture be developed, measured, validated, and progressively improved as an integrated performance-regulated system?

Research Method

The Lab organizes research through the sequence: Foundation → Unit → Measurement → System → Validation → Scale. Research moves from conceptual foundations to operational units, measurable indicators, integrated systems, empirical validation, and scalable application.

Lines of Inquiry

Energy and Environmental Regulation Systems — principles governing exchanges of energy, water, and matter between built environments and their climatic context.

Metabolic Architecture — the PRMS node as a minimum functional unit defined by producing, regulating, inhabiting, and measuring.

Performance Measurement — composite indicators for verifiable building performance, including the PRMS Score.

Data-Driven Systems — data acquisition, processing, feedback, and decision architectures for monitored built environments.

Applied Prototyping — construction and post-occupancy evaluation of physical nodes and instrumented buildings as experimental environments.

Urban Metabolism and Distributed Urban Systems — aggregation from dwelling to district and the emergence of collective resource-flow and resilience properties.

Research Programs and Infrastructures

PRMS Framework — conceptual and methodological formalization.

PRMS Score — measurement and evaluation infrastructure for metabolic performance.

SmartLiving Milenio III — applied living-lab and prototype validation environment.

PRMS AI — emerging research infrastructure for architectural intelligence, discovery, simulation, experimentation, and knowledge accumulation.

Renewable Horizons — editorial and conceptual research platform supporting the wider research corpus.

Evidence and Validation

Research should distinguish evidence according to its origin and reliability, including observed, declared, inferred, simulated, and verified information where applicable. Measured performance, documented projects, experimental results, research publications, and persistent research records constitute the principal evidence streams.

2026 Development Priorities

Formalize and consolidate the institutional research record; strengthen persistent scholarly records; advance empirical validation of PRMS systems and performance measurement; develop cumulative data, knowledge, and AI-assisted discovery infrastructure; strengthen interdisciplinary dialogue and collaboration; prepare the institutional evidence base required for future organizational identifiers and broader scholarly integration.

Expected Research Outputs

Research articles and preprints; books and research documents; taxonomies, frameworks, and methodological protocols; measured project records; performance evaluations; experimental results; software and technical infrastructure; and validated knowledge that can inform subsequent research.

Agenda Governance

This is a living institutional document. Significant changes to research scope, major research lines, or institutional structure should be documented through a new version.